



Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: None (Standard Ubuntu Server)
3. Wait for installation to complete and

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1:** Enable Network Adapter
4. **Attached to:** Select **Bridged Adapter**
5. **Name:** Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode
4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` – installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtproto kernel_rdr proto kernel_ll
        valid_lft 179068sec preferred_lft 83386sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft 179068sec preferred_lft 92668sec
    valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:                0.0
Usage of /:                  5.1% of 48.91GB
Memory usage:               5%
Swap usage:                 0%
Processes:                  139
Users logged in:            1
IPv4 address for enp0s3:    192.168.1.14
IPv6 address for enp0s3:    2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots

(labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSCS – 3

Date Submitted: September 3, 2026

Instructor: Johneros P. Puyo

1. OBJECTIVES (5 POINTS) Check all that apply:

- ☒ Create and configure a Virtual Machine with Ubuntu Server
- ☒ Configure Bridged Network Adapter for network access
- ☒ Determine and verify the server's IP address
- ☒ Install OpenSSH Server and enable the service
- ☒ Establish successful SSH connection from host machine
- ☒ Execute verification commands to confirm remote access

2. HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu Server OS
Virtualization Software (VirtualBox/VMware)	Oracle VM VirtualBox
Ubuntu Server Version	ubuntu-26.04-live-server-amd64
VM Name	UBUNTU-SERVER
RAM Allocated	8192 MB 8 GB
Storage Allocated	100 GB
Network Mode	Bridged Adapter
SSH Client Used	OpenSSH

3. VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	<u>8192</u> MB
Hard Disk	<u>100</u> GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION Commands Executed:

Server IP Address: 169.254.175.61

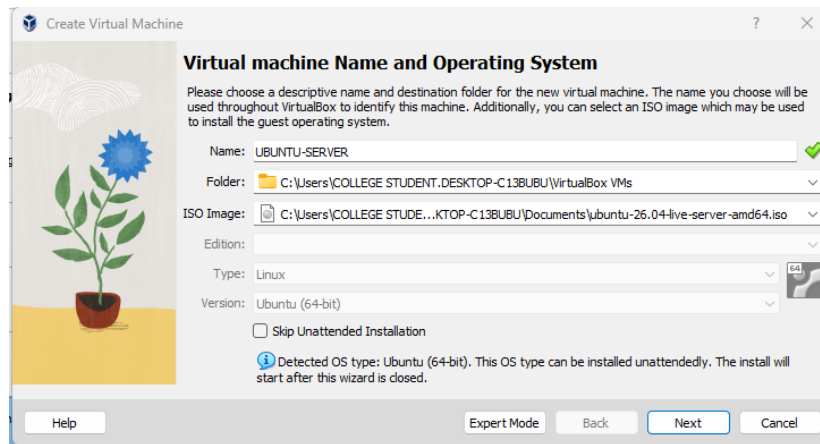
Network Interface: Physical

5. REMOTE CONNECTION Connection Details:

Parameter	Value
Server IP	169.254.175.61
Username	isotropic
SSH Port	22

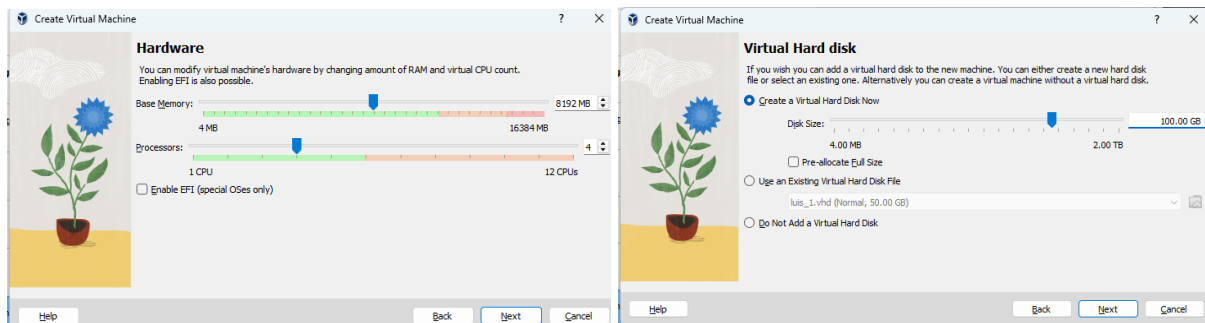
SETUP PROCESS

1. Click the “New” button on the Virtual Box and this window will pop out.

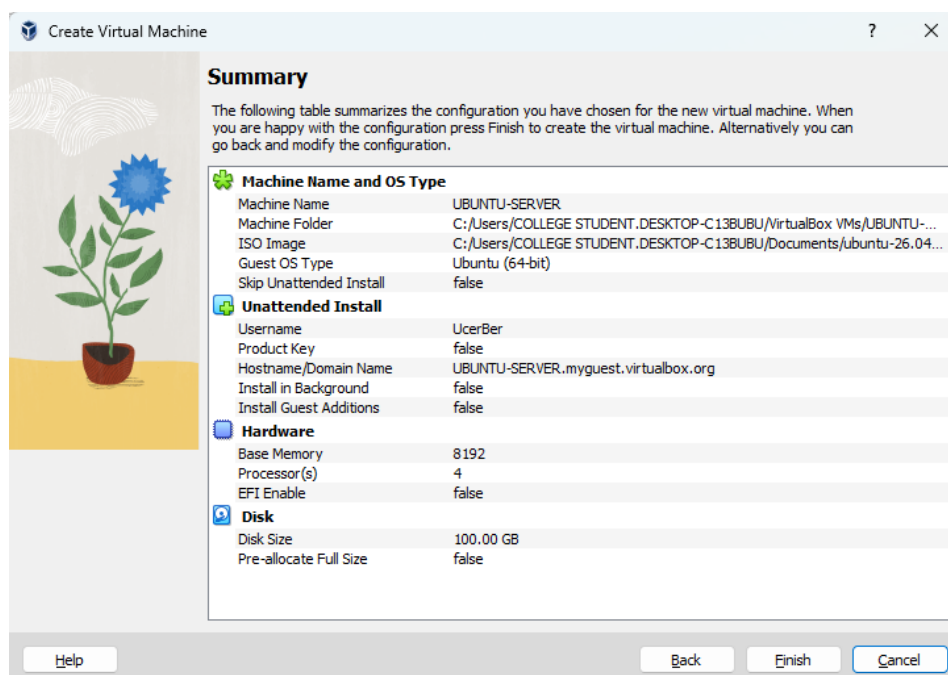


Give your VM Server a name and select the chosen OS, for this one we will choose the ubuntu live server.

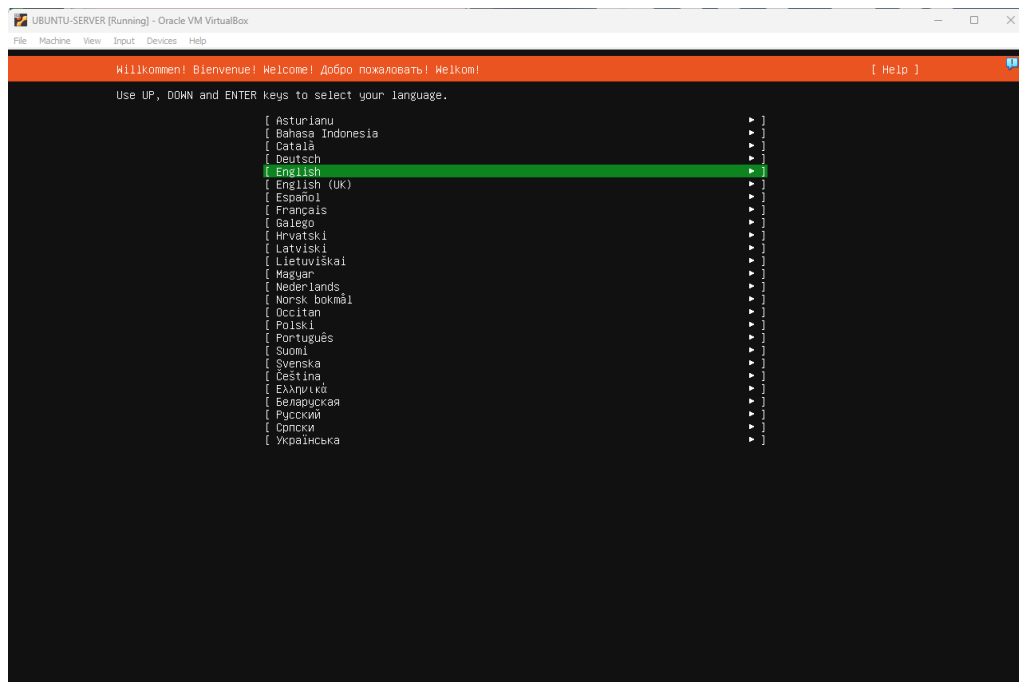
2. Allocate the standard Size for Memory, CPU Cores, and Storage.



3. Preview the Summary of the VM Specification and click FINISH.

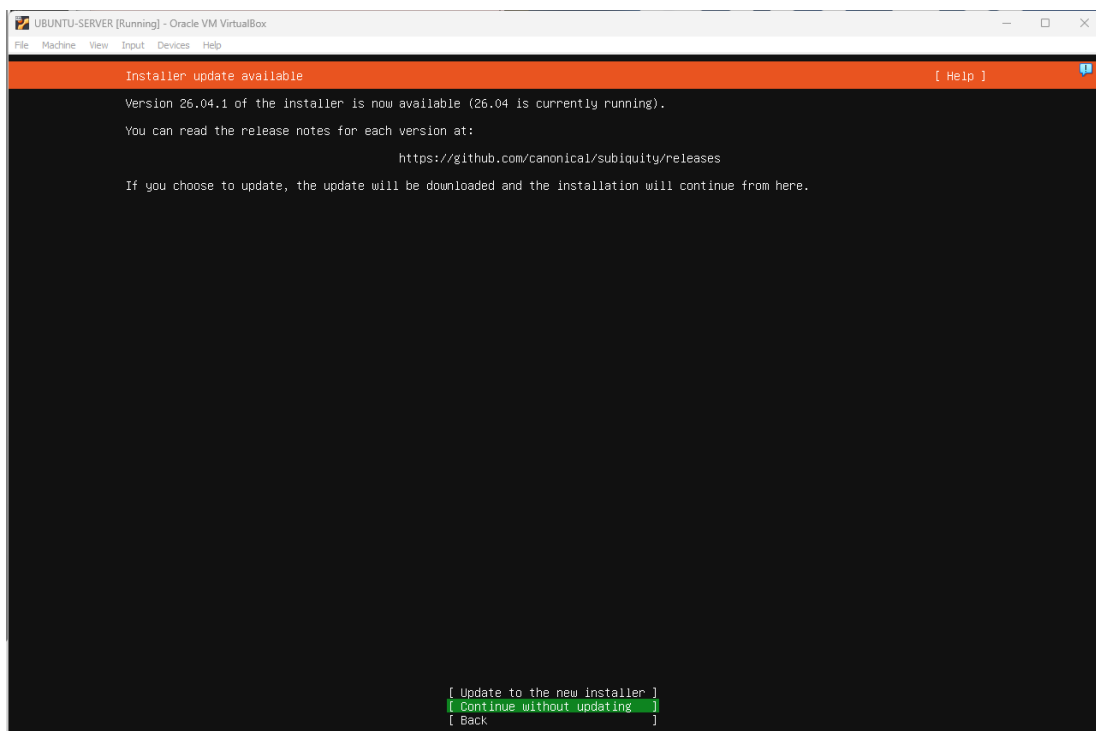


4. After waiting for a while, the Virtual Machine will automatically Opened. From here the only actions you can do is Up and Down buttons and for selection/confirmation is the Enter key.

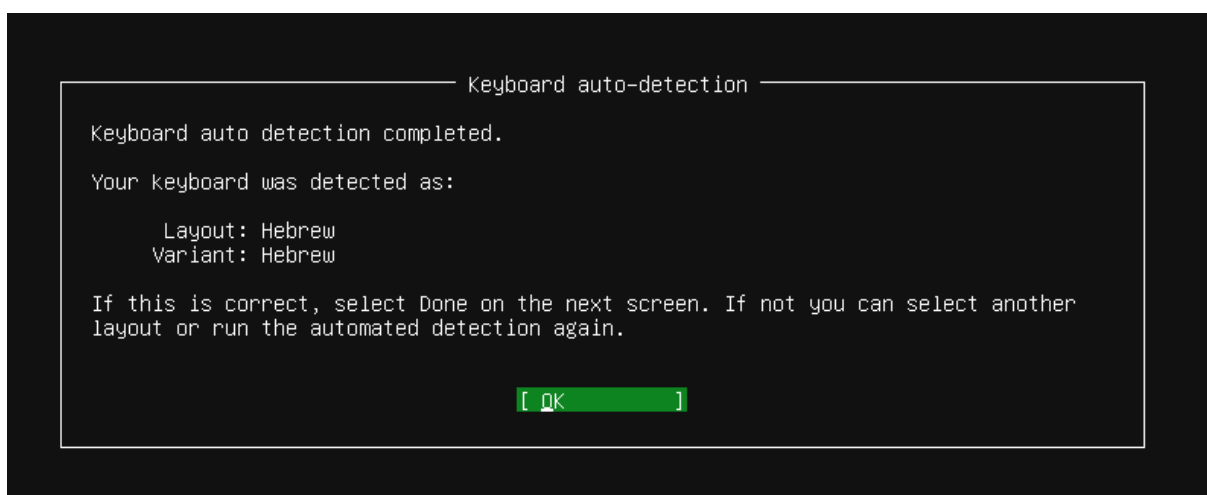
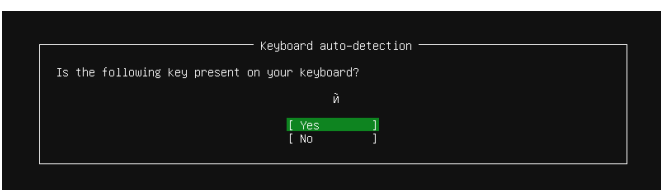
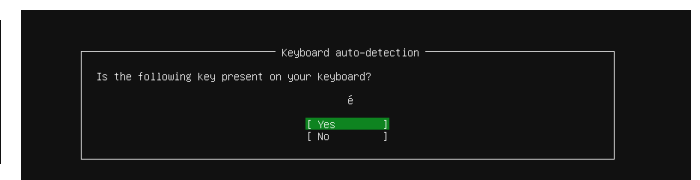
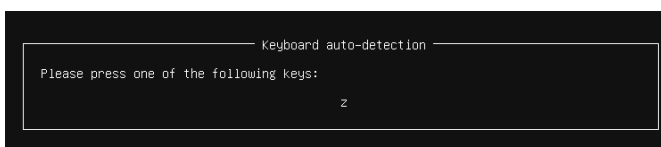
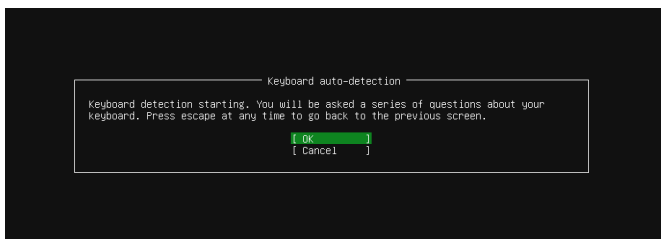
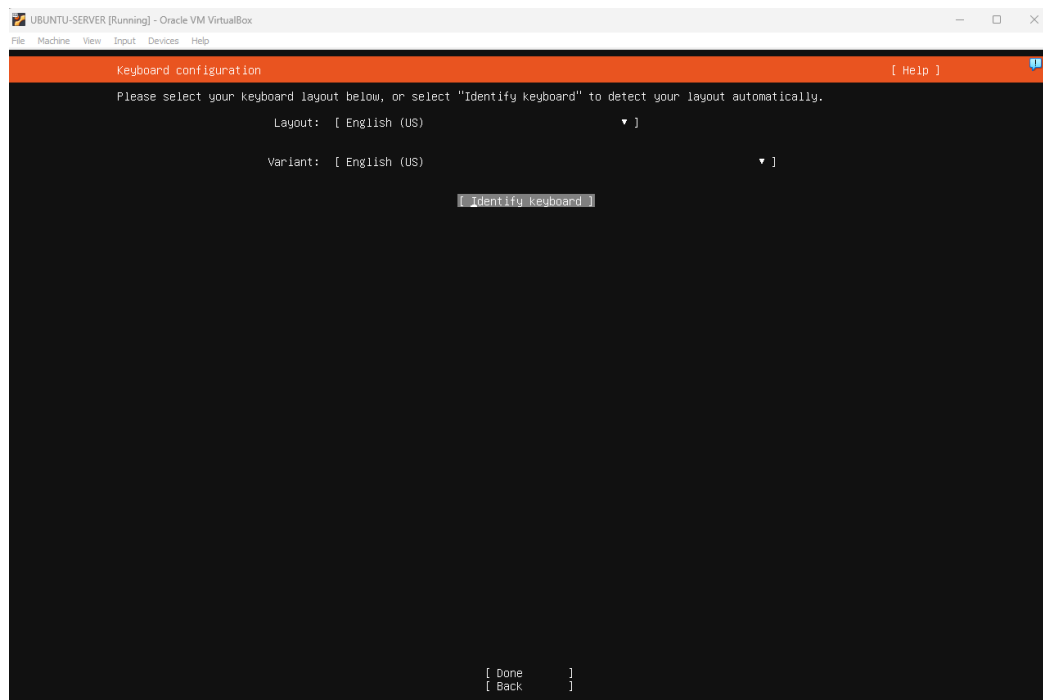


Here, you can choose whatever language you preferred.

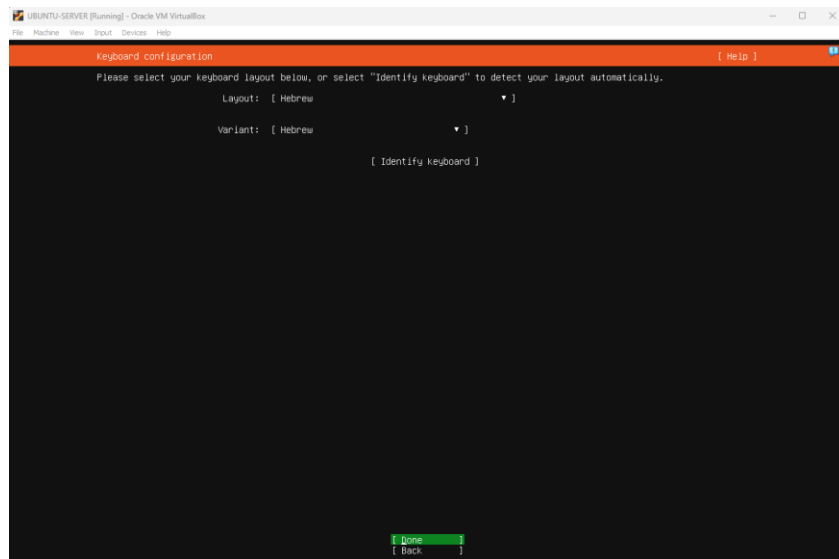
5. Then you can choose whether you want the updated version of the OS or not.



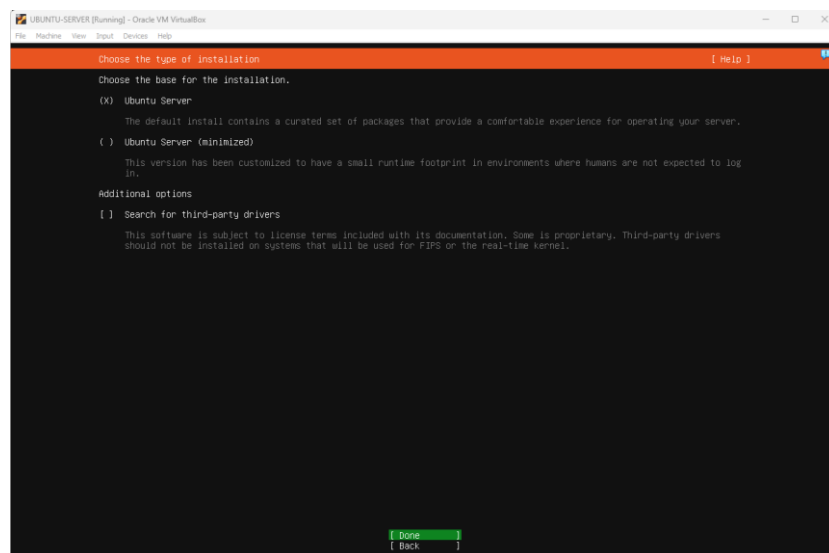
6. Here, I use the auto detection for the keyboard.



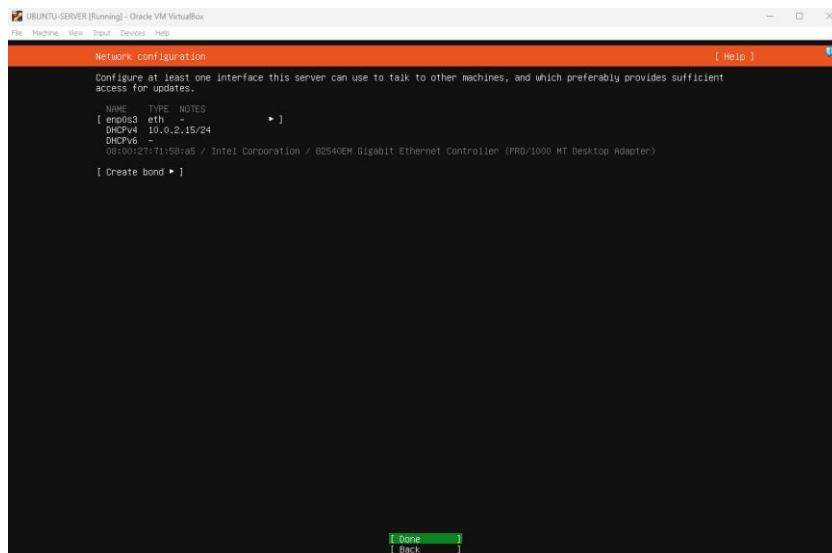
Here, the auto detection is done.



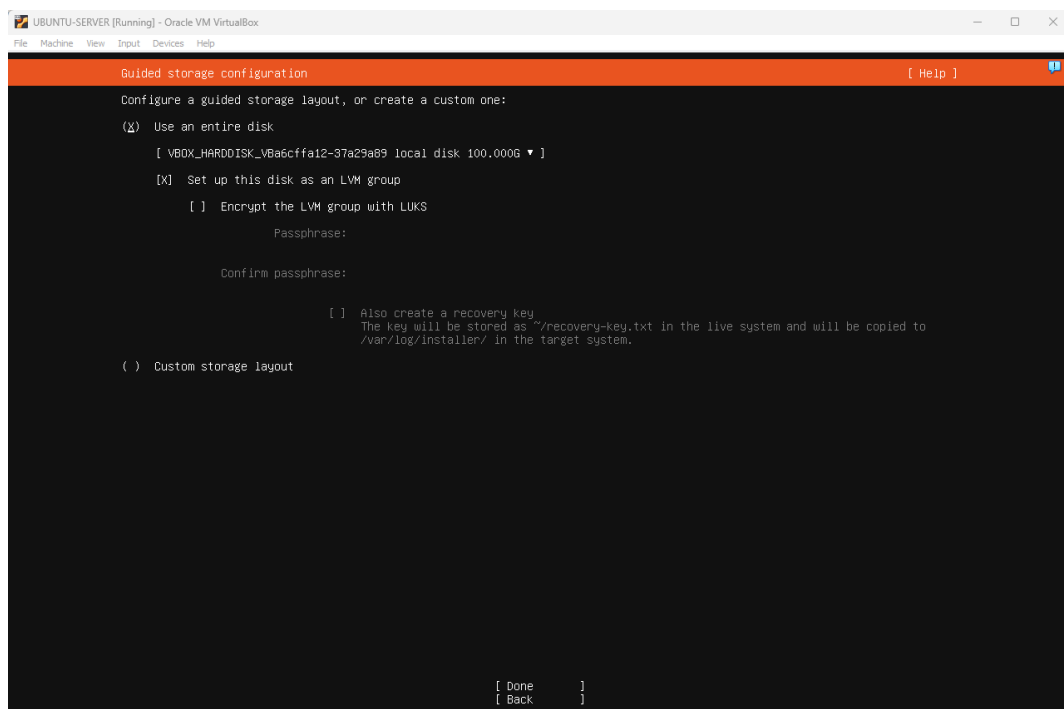
7. For the type of Installation, I just choose the “Ubuntu Server”



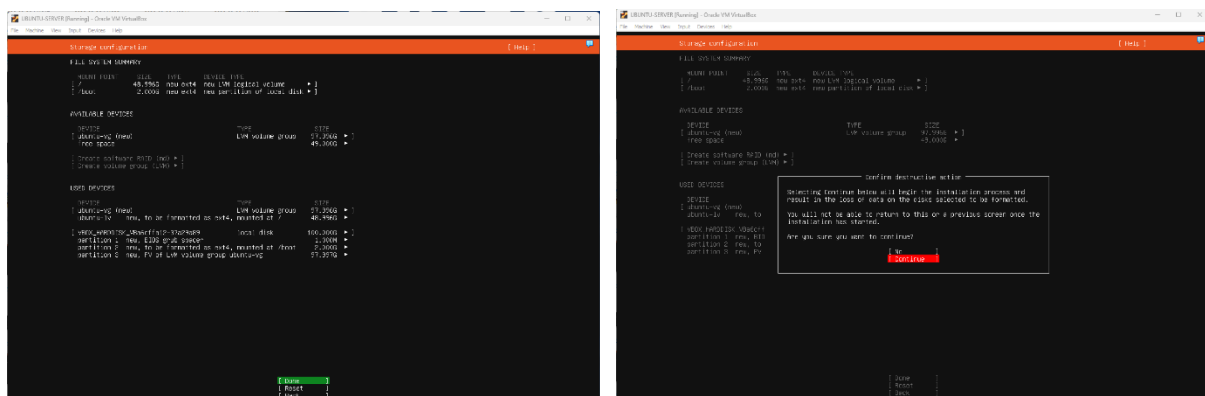
8. For network config, we will not touch anything here for now and use the defaults.



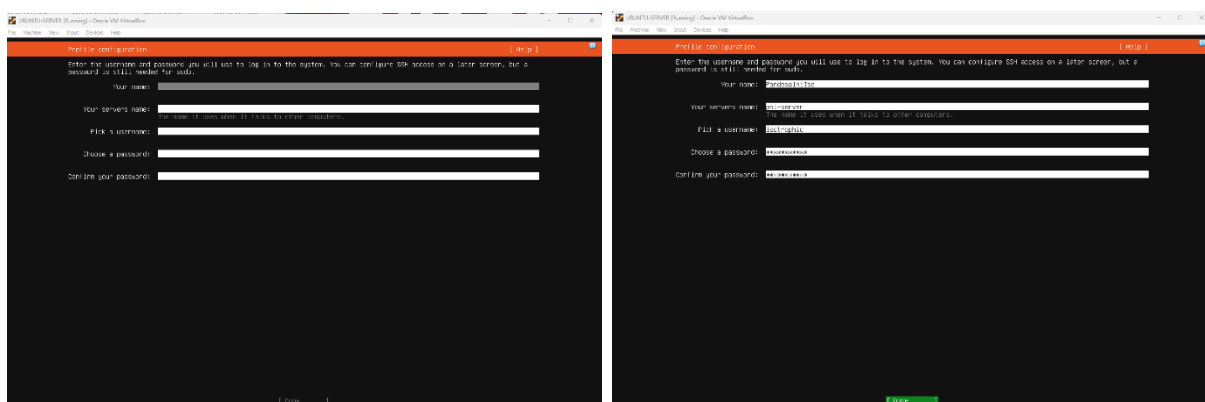
9. For the Storage Allocations, we will use the entire capacity.



10. Since we just choose the entire disk to be use, it automatically distributes to each partitions.

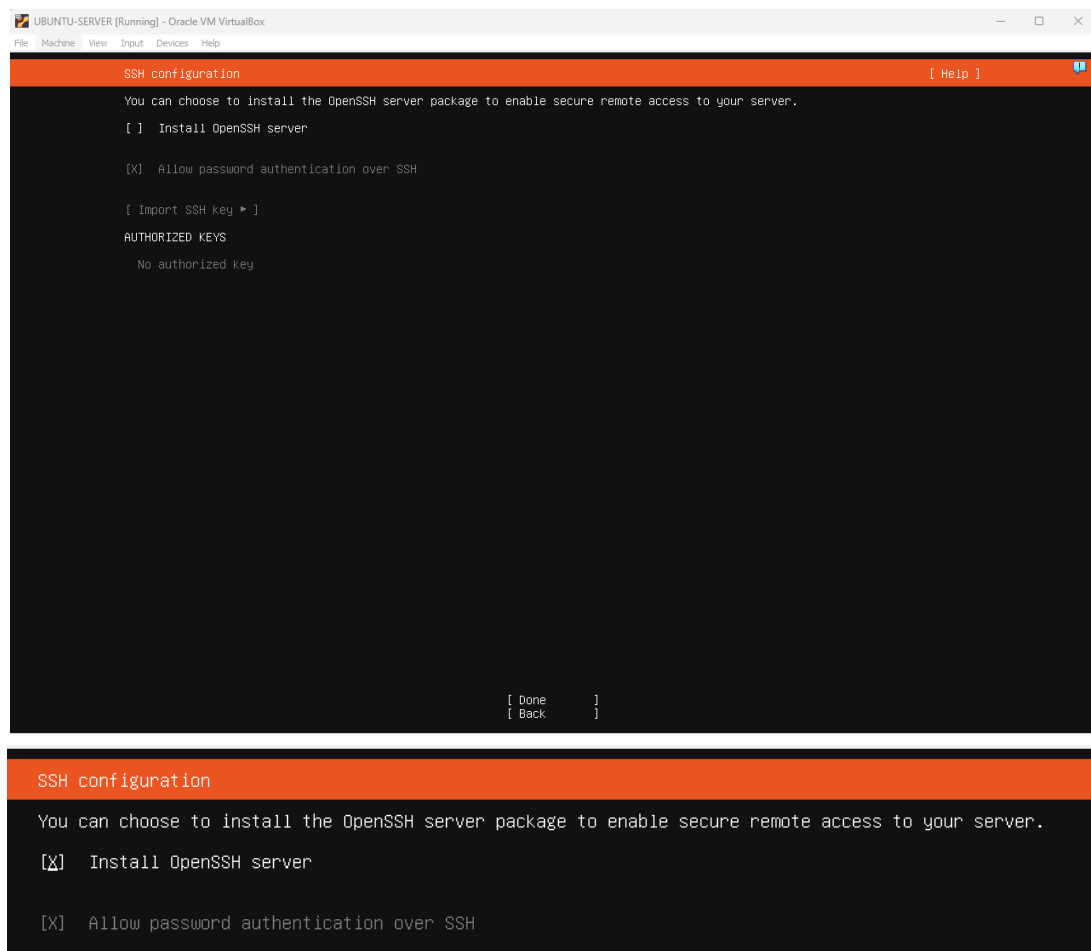


11. For Profile Config, put your desired details for your Ubuntu Server.

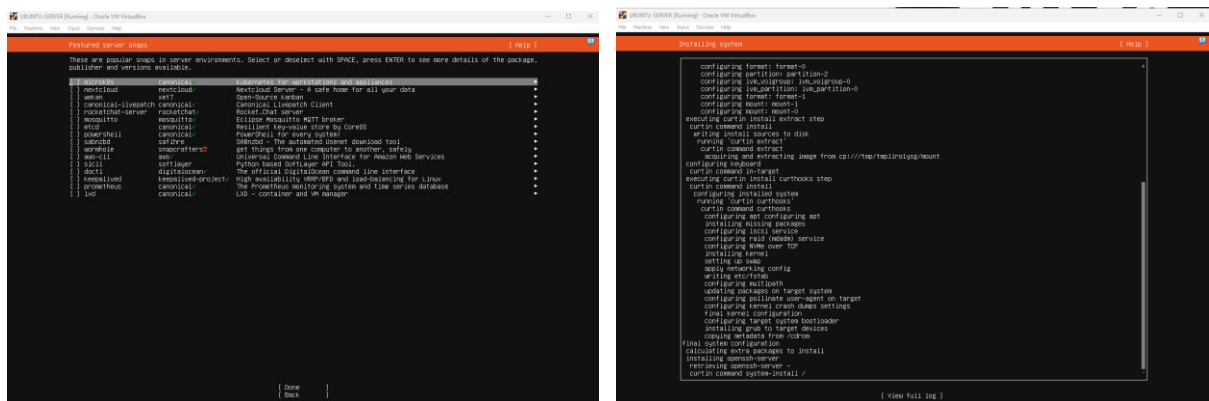


Once your satisfied to your preferred details, just click Done.

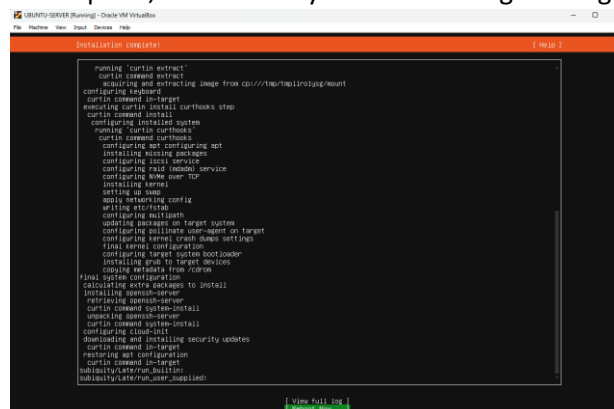
12. For SSH config, choose the Install OpenSSH server so we can access this to our client VM.



13. Here, you don't have to choose any of there, after selecting Done the installation of the Ubuntu Server will start.



14. Installation Progress is Complete, reboot the system and we'll good to go.



15. Here's the look of freshly installed Ubuntu Server.

```
UBUNTU-SERVER [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
13.248943 cloud-init[1486]: The key's randomart image is:
13.248976 cloud-init[1486]: +---[RSA 3072]----+
13.249063 cloud-init[1486]: |... 0... .0...==+
13.249099 cloud-init[1486]: |..+ .....0+0+0
13.249131 cloud-init[1486]: |..+ .....+0=
13.249160 cloud-init[1486]: |.. 0...++
13.249183 cloud-init[1486]: |.. S 0...+
13.249211 cloud-init[1486]: |.. .+ 0..0..
13.249236 cloud-init[1486]: |.. + ..+..0
13.249261 cloud-init[1486]: |E .. 0..0.0
13.249289 cloud-init[1486]: |.. .?....+
13.249313 cloud-init[1486]: +---[SHA256]-----+
13.249330 cloud-init[1486]: Generating public/private ecdsa key pair.
13.249390 cloud-init[1486]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
13.249436 cloud-init[1486]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
13.249485 cloud-init[1486]: The key fingerprint is:
13.249512 cloud-init[1486]: SHA256:3hA07pvrC7fbISD0X6V5nK9nItYduYos2kUDSx00vSa root@pni-server
13.249546 cloud-init[1486]: The key's randomart image is:
13.249574 cloud-init[1486]: +---[ECDSA 256]---+
13.249600 cloud-init[1486]: |.. 0
13.249611 cloud-init[1486]: |.. + 0
13.249667 cloud-init[1486]: |..Eoo.
13.249697 cloud-init[1486]: |..+0+0
13.249721 cloud-init[1486]: |.. 0 S..
13.249742 cloud-init[1486]: |.. + +
13.249764 cloud-init[1486]: |.. 0 0 +0...
13.249787 cloud-init[1486]: |.. .+00* 0..
13.249806 cloud-init[1486]: |.. .+0+0B800.
13.249829 cloud-init[1486]: +---[SHA256]-----+
13.249881 cloud-init[1486]: Generating public/private ed25519 key pair.
13.249910 cloud-init[1486]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
13.249939 cloud-init[1486]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
13.249966 cloud-init[1486]: The key fingerprint is:
13.249994 cloud-init[1486]: SHA256:1Q7rrhJID7hucE4JLx2TpMupVCU03gmfxAJYSxfd2g root@pni-server
13.250006 cloud-init[1486]: The key's randomart image is:
13.250113 cloud-init[1486]: +---[ED25519 256]---+
13.250135 cloud-init[1486]: |====..0.
13.250156 cloud-init[1486]: |B=0..E.
13.250177 cloud-init[1486]: |B+=
13.250199 cloud-init[1486]: |0+000..
13.250220 cloud-init[1486]: |=-0.B.S
13.250242 cloud-init[1486]: |=-0.B=
13.250264 cloud-init[1486]: |.=+ +.
13.250287 cloud-init[1486]: |...0
13.250309 cloud-init[1486]: |...00
13.250329 cloud-init[1486]: +---[SHA256]-----+
ubuntu 26.04 LTS pni-server tty1
pni-server login:
pni-server login: isotrophic
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

+ Documentation:  https://docs.ubuntu.com
+ Management:    https://landscape.canonical.com
+ Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 02:55:55 AM UTC 2026

System load:  0.83          Processes:      157
Usage of /:   15.0% of 47.9GB Users logged in:    0
Memory usage: 3%           IPV4 address for enp0s3: 10.0.2.15
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

55 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

isotrophic@pni-server:~$
```

Here what it look's like after logging in to our Server

```
UBUNTU-SERVER [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
13.249966 cloud-init[1486]: The key fingerprint is:
13.249994 cloud-init[1486]: SHA256:1Q7rrhJID7hucE4JLx2TpMupVCU03gmfxAJYSxfd2g root@pni-server
13.250006 cloud-init[1486]: The key's randomart image is:
13.250113 cloud-init[1486]: +---[ED25519 256]---+
13.250135 cloud-init[1486]: |====..0.
13.250156 cloud-init[1486]: |B=0..E.
13.250177 cloud-init[1486]: |B+=
13.250199 cloud-init[1486]: |0+000..
13.250220 cloud-init[1486]: |=-0.B.S
13.250242 cloud-init[1486]: |=-0.B=
13.250264 cloud-init[1486]: |.=+ +.
13.250287 cloud-init[1486]: |...0
13.250309 cloud-init[1486]: |...00
13.250329 cloud-init[1486]: +---[SHA256]-----+
ubuntu 26.04 LTS pni-server tty1
pni-server login: isotrophic
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

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individual files in /usr/share/doc/*/copyright.

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applicable law.

isotrophic@pni-server:~$
```

16. From here, let's install some additional resources for SSH

- `sudo systemctl status ssh`

```
isotrophic@pni-server:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 02:55:06 UTC; 1min 53s ago
   Invocation: c3da72f7c5c94785884ecfd5bf240ecf
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 5566 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 5569 (sshd)
     Tasks: 1 (limit: 6192)
   Memory: 1.6M (peak: 2.4M)
     CPU: 19ms
   CGroup: /system.slice/ssh.service
           └─5569 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 03 02:55:06 pni-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 03 02:55:06 pni-server sshd[5569]: Server listening on 0.0.0.0 port 22.
Sep 03 02:55:06 pni-server sshd[5569]: Server listening on :: port 22.
Sep 03 02:55:06 pni-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

- *sudo apt update*

```
isotrophic@pni-server:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
58 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

- *sudo apt install openssh-server* (if ever you missed to check the installation of OpenSSH)

```
isotrophic@pni-server:~$ sudo apt install openssh-server
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 58
```

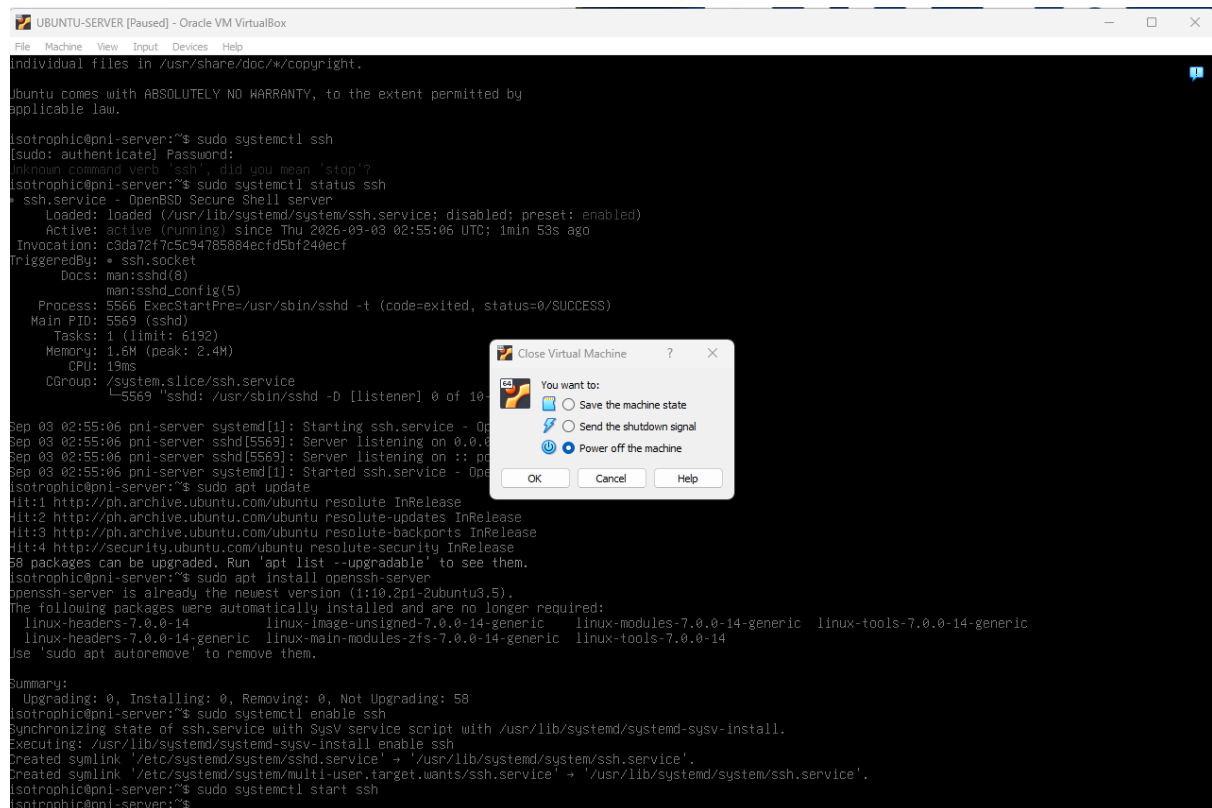
- *sudo systemctl enable ssh*

```
isotrophic@pni-server:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' + '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' + '/usr/lib/systemd/system/ssh.service'.
```

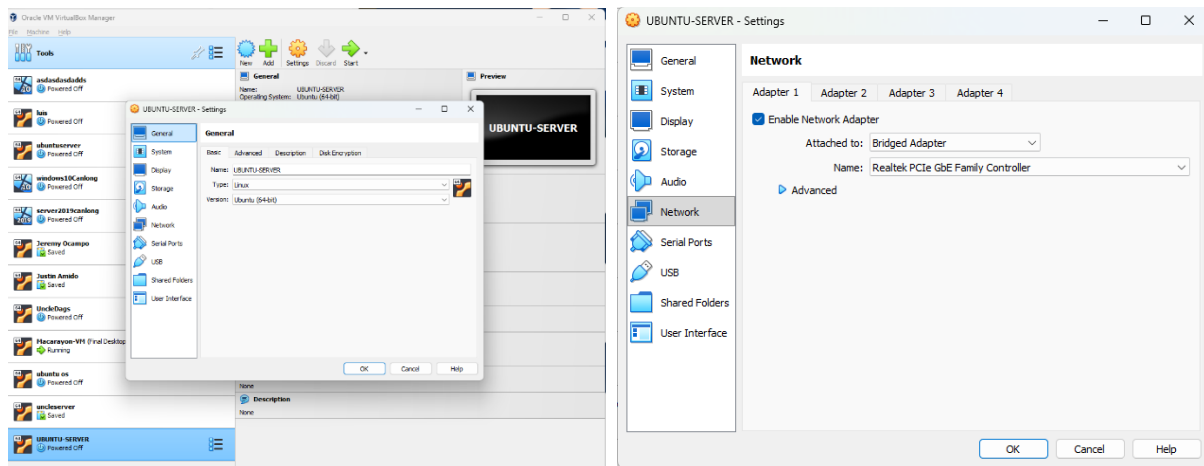
- *sudo systemctl start ssh*

```
isotrophic@pni-server:~$ sudo systemctl start ssh
isotrophic@pni-server:~$
```

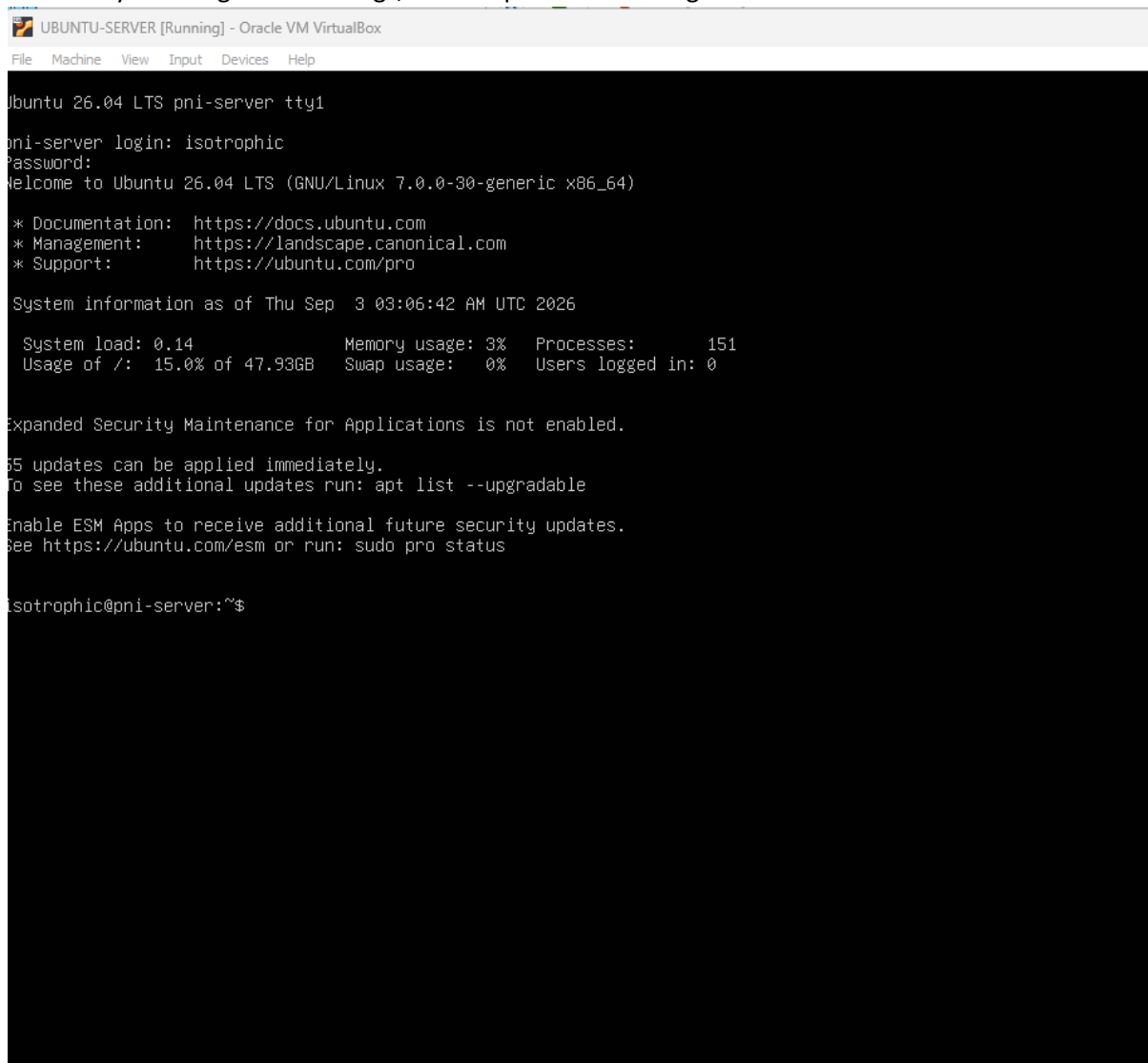
Once everything is installed, we will shutdown the VM



17. After Shutting down, go for the setting of the VM server, change “attached to” from NAT into Bridged Adapter.



18. Once you changed the settings, Power up the machine again



19. From here, if we look for our IP details under the “enp0s3” we don’t have a IPv4 Address.

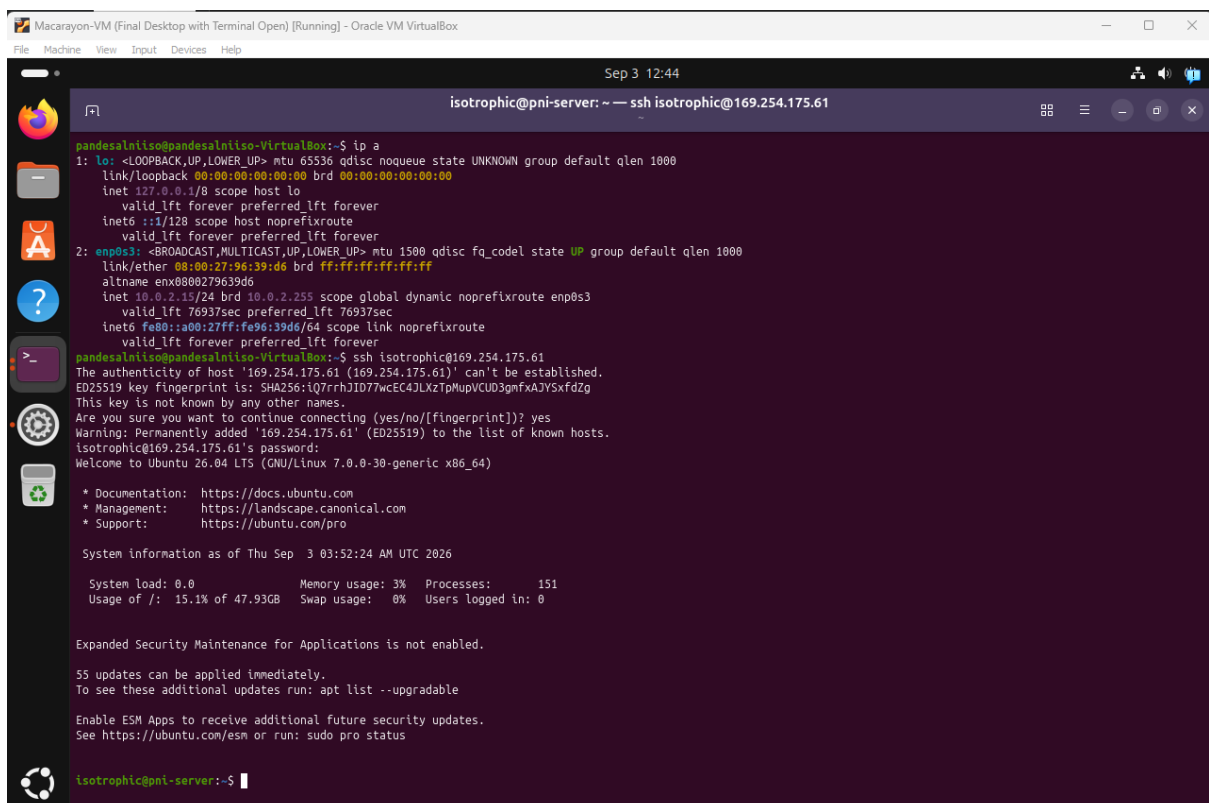
- ip a

```
isotrophic@pni-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:71:58:a5 brd ff:ff:ff:ff:ff:ff
    altname enx0800277158a5
    inet6 fe80::a00:27ff:fe71:58a5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
isotrophic@pni-server:~$
```

To get an IP address, type “sudo dhcpcd”, then to check if it's applied, type “ip a” again.

```
isotrophic@pni-server:~$ sudo dhcpcd
[sudo: authenticate] Password:
dhcpcd-10.3.0 starting
dev: loaded udev
DUID 00:01:00:01:32:2b:a9:2b:08:00:27:71:58:a5
enp0s3: IAID 27:71:58:a5
enp0s3: soliciting an IPv6 router
enp0s3: soliciting a DHCP lease
enp0s3: probing for an IPv4LL address
enp0s3: using IPv4LL address 169.254.175.61
enp0s3: adding route to 169.254.0.0/16
enp0s3: adding default route
Dropped protocol specifier '.ipv4ll' from 'enp0s3.ipv4ll'. Using 'enp0s3' (ifindex=2).
isotrophic@pni-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:71:58:a5 brd ff:ff:ff:ff:ff:ff
    altname enx0800277158a5
    inet 169.254.175.61/16 brd 169.254.255.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe71:58a5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

20. Once the IP Address is confirmed, we can go to our Client VM to SSH it



Macarony-VM (Final Desktop with Terminal Open) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Sep 3 12:44

isotrophic@pni-server: ~ — ssh isotrophic@169.254.175.61

```
pandesalnis@pandesalnis-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:96:39:d6 brd ff:ff:ff:ff:ff:ff
    altname enx0800279639d6
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 76937sec preferred_lft 76937sec
    inet6 fe80::a00:27ff:fe96:39d6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
pandesalnis@pandesalnis-VirtualBox:~$ ssh isotrophic@169.254.175.61
The authenticity of host '169.254.175.61 (169.254.175.61)' can't be established.
ED25519 key fingerprint is: SHA256:iQ7rrhJI077wCEC4JLxZIpMupVCUD3gnfxAJVSxfdzg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '169.254.175.61' (ED25519) to the list of known hosts.
isotrophic@169.254.175.61's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 03:52:24 AM UTC 2026

System load: 0.0          Memory usage: 3%          Processes:   151
Usage of /: 15.1% of 47.93GB  Swap usage:  0%          Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

55 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

isotrophic@pni-server:~$
```

- ssh [username of server]@[IP Address of server] | ssh isotrophic @169.254.175.61

```
pandesalniiso@pandesalniiso-VirtualBox:~$ ssh isotrophic@169.254.175.61
The authenticity of host '169.254.175.61 (169.254.175.61)' can't be established.
ED25519 key fingerprint is: SHA256:iQ7rrhJID77wcEC4JLXzTpMupVCUD3gmfxAJYSxfdZg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '169.254.175.61' (ED25519) to the list of known hosts
.
isotrophic@169.254.175.61's password:
```

To get access to our server, type the server's password.

```
isotrophic@pni-server: ~ — ssh isotrophic@169.254.1...
.
isotrophic@169.254.175.61's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

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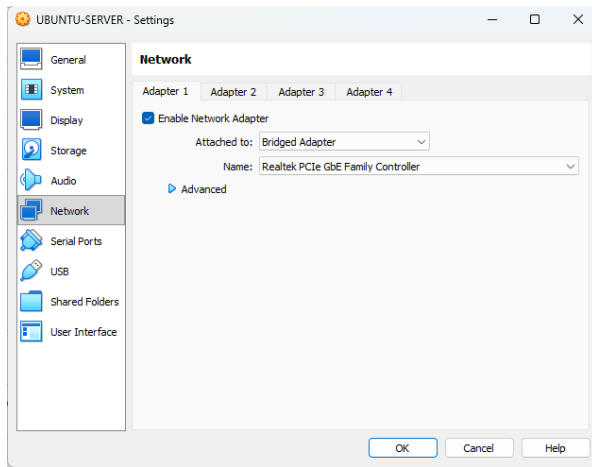
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

isotrophic@pni-server:~$
```

And now we're done — we've successfully SSH'd into our server.

Required Screenshots

VM Network Settings



Server IP Address

```
isotrophic@pni-server:~$ sudo dhcpcd
[sudo: authenticate] Password:
dhcpcd-10.3.0 starting
dev: loaded udev
UID 00:01:00:01:32:2b:a9:2b:08:00:27:71:58:a5
enp0s3: IAD 27:71:58:a5
enp0s3: soliciting an IPv6 router
enp0s3: soliciting a DHCP lease
enp0s3: probing for an IPv4LL address
enp0s3: using IPv4LL address 169.254.175.61
enp0s3: adding route to 169.254.0.0/16
enp0s3: adding default route
Dropped protocol specifier '.ipv4ll' from 'enp0s3.ipv4ll'. Using 'enp0s3' (ifindex=2).
isotrophic@pni-server:~$ ip a
1: lo: loopback UP,LOOPUP, mtu 65536 qdisc noop state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: GROUNDCAST, MULTICAST, UP, LOWER_UP, mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff
    altname enx000000000000
    inet 169.254.175.61/16 brd 169.254.255.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe11:58a5/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
```

Before SSH Access Status

```
isotrophic@pni-server:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 02:55:06 UTC; 1min 53s ago
   Invocation: c3da72f7c5c94785884ecfd5bf240ecf
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
         man:sshd_config(5)
   Process: 5566 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 5569 (sshd)
     Tasks: 1 (limit: 6192)
    Memory: 1.6M (peak: 2.4M)
       CPU: 19ms
   CGroup: /system.slice/ssh.service
           └─5569 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

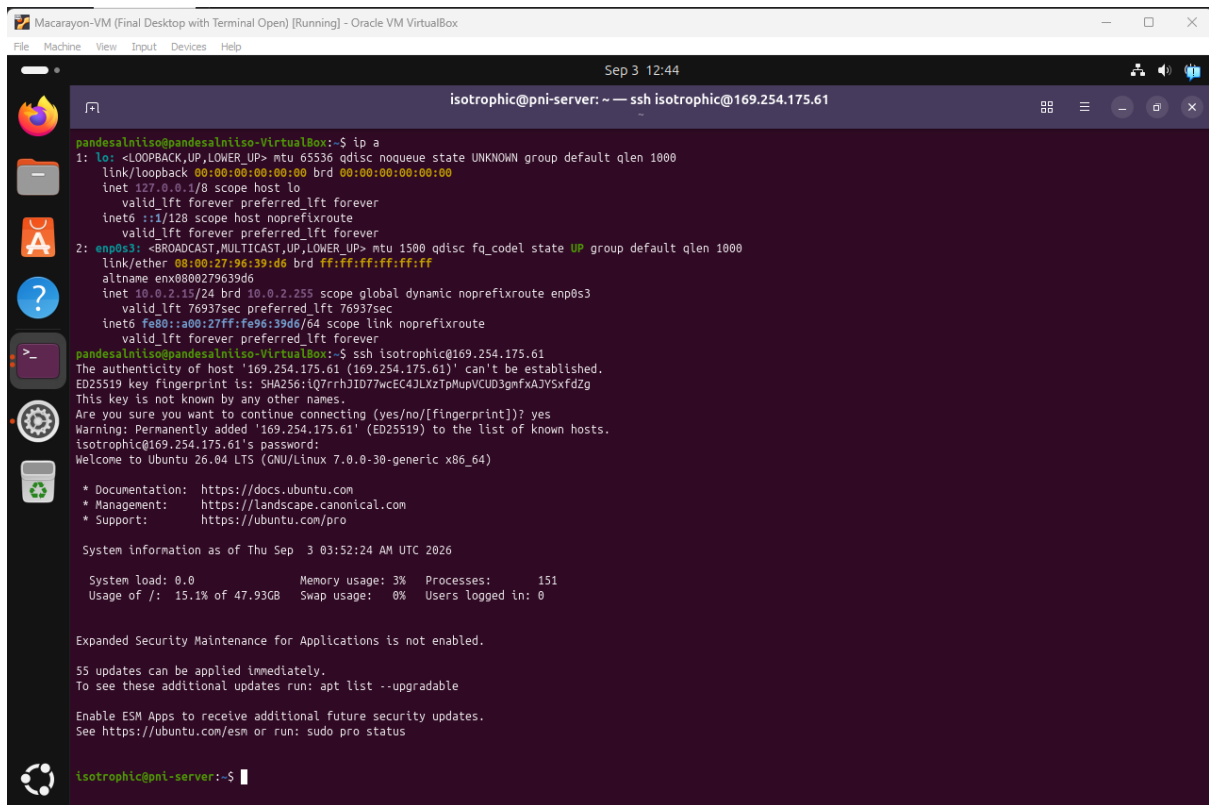
Sep 03 02:55:06 pni-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server..
Sep 03 02:55:06 pni-server sshd[5569]: Server listening on 0.0.0.0 port 22.
Sep 03 02:55:06 pni-server sshd[5569]: Server listening on :: port 22.
Sep 03 02:55:06 pni-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

SSH Access Status

```
isotrophic@pni-server:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 03:40:43 UTC; 19min ago
   Invocation: 28c0d147f304409fbee57e4b89a62e0a
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
         man:sshd_config(5)
   Process: 1325 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1333 (sshd)
     Tasks: 1 (limit: 6192)
    Memory: 6.6M (peak: 9.1M)
       CPU: 57ms
   CGroup: /system.slice/ssh.service
           └─1333 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 03 03:40:43 pni-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 03 03:40:43 pni-server sshd[1333]: Server listening on 0.0.0.0 port 22.
Sep 03 03:40:43 pni-server sshd[1333]: Server listening on :: port 22.
Sep 03 03:40:43 pni-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Sep 03 03:52:24 pni-server sshd-session[1594]: Accepted password for isotrophic from 172.16.18.20 port 51563 ssh2
Sep 03 03:52:24 pni-server sshd-session[1594]: pam_unix(sshd:session): session opened for user isotrophic(uid=1000) by isotrophic(uid=0)
```

Successful SSH connection from host



The screenshot shows a terminal window titled "Macaroyon-VM (Final Desktop with Terminal Open) [Running] - Oracle VM VirtualBox". The terminal is running on a host named "isotrophic@pni-server" with IP 169.254.175.61. The user "pandesalntiiso" is logged into a virtual machine named "pandesalntiiso-VirtualBox".

The terminal output shows the following commands and results:

```
pandesalntiiso@pandesalntiiso-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:96:39:d6 brd ff:ff:ff:ff:ff:ff
    altname enx0800279639d6
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 76937sec preferred_lft 76937sec
    inet6 fe80::a00:27ff:fe96:39d6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
pandesalntiiso@pandesalntiiso-VirtualBox:~$ ssh isotrophic@169.254.175.61
The authenticity of host '169.254.175.61 (169.254.175.61)' can't be established.
ED25519 key fingerprint is: SHA256:q7rrhJID77wCEC4JLXzTpMupVCUD3gmfxAJY5xZdZg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '169.254.175.61' (ED25519) to the list of known hosts.
isotrophic@169.254.175.61's password:
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isotrophic@pni-server:~$
```